



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013
November
Year 11 Assessment
Task 1

Mathematics (2 Unit)

General Instructions

- Reading Time – 5 Minutes
- Working time – 1 ½ Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may NOT be awarded for messy or badly arranged work.
- EACH QUESTION IS TO BE RETURNED IN A SEPARATE BUNDLE. Q1, Q2, Q3, Q4 and Q5
- All necessary working should be shown in every question if full marks are to be awarded
- Leave all answers in simplified exact form unless indicated otherwise.

Total Marks – 80

- Attempt questions 1 – 5

Examiner: *F Nesbitt*

Question 1 (16 marks)

- (a) Show that $x^4 - 3x^2 - 2$ is an even function. 2
- (b) Find $\lim_{x \rightarrow 3} \frac{3x}{x+2}$ 1
- (c) Evaluate $\sum_{n=1}^6 (2n-3)$ 2
- (d) Differentiate:
- (i) $y = x^2 + 2x$
- (ii) $y = 3x - \frac{2}{x^3}$
- (iii) $y = \frac{2x}{1+x^2}$ 5
- (e) Simplify fully:
- (i) $a^5 \div a^{-3}$
- (ii) $\left(\frac{2n^2}{-m}\right)^3 \left(\frac{-m^3}{n}\right)^2$ 3
- (iii) $\log_3 81 + \log_3 9 + \log_3 243$ 1
- (f) $f(x) = 3x^3 - 3x^2 + bx + 5$
- Find the value of b if $f(x)$ has only one stationary point. 2

Question 2: (16 marks)

(a) The quadratic equation $3x^2 + 9x + 1 = 0$ has roots α and β . Write the value of:

(i) $\alpha + \beta$

(ii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iii) $(\alpha + 2)(\beta + 2)$ 5

(b) Find the equation of the normal to $y = \frac{x^2}{2}$ at the point (4,8). 3

(c) Find the largest possible domain and range for each of the following functions

(i) $y = 3x^2 - 5$

(ii) $y = 3^x + 1$

(iii) $y = \frac{1}{x+5}$

(iv) $y = \frac{1}{\sqrt{2-x^2}}$ 8

Question 3 (16 marks)

- (a) A die is biased so that to throw a six is twice as likely as throwing any other number. What is the probability of throwing:
- (i) a two?
 - (ii) a six? 2
- (b) The first four terms of an arithmetic series are $-5 -1+ 3+ 7 +\dots$
- (i) What is the n th term
 - (ii) Find the sum of the first 20 terms 3
- (c) Find the equation of the locus of a point which is always $\sqrt{3}$ units From the point $(-3, 5)$ 1
- (d) Find the equation of the locus of a point that moves so that It is always equidistant from the line $6x + 8y - 5 = 0$ and The line $5x + 12y - 1 = 0$. 3
- (e) Lily bought 2 tickets in a small raffle. 20 tickets were sold. There were two prizes. What is the probability that Lily won:
- (i) both prizes?
 - (ii) at least one prize? 3
- (f) Find what value(s) of k does the equation $x^2 + kx + 8 - k = 0$ have
- (i) equal roots (ii) real and distinct roots 4

Question 4 (16 marks)

- (a) Express the following series in sigma notation:

$$1 \times 4 + 2 \times 5 + 3 \times 6 + \dots + 10 \times 13 \quad 2$$

- (b) What is the value of $f(3a - 1)$ if $f(x) = (2x - 3)$? 2

- (c) Simplify fully:

(i) $8^{\frac{1}{2}} \div 2^{\frac{1}{2}}$

(ii) $\log_a(a^2 + a) - \log_a(a + 1)$ 3

- (d) For the curve $f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x - 1$:

- (i) Find the coordinates of any stationary points and determine their nature. 3

- (ii) Find any point(s) of inflexion. 2

- (iii) Sketch the curve in the domain $-4 \leq x \leq 3$ 3.

- (iv) What is the maximum value of $f(x)$ in the given domain? 1

Question 5 (16 marks)

- (a) Write the coordinates of the focus and the equation of the directrix of the parabola $3y^2 = 4x$ 2
- (a) Solve the inequation: $2x^2 - 9x + 7 > 0$ 3
- (b) Find, from first principles, the derivative of $y = x^2 - 6x + 5$ 3
- (c) Find the domain and range of $y = \log(x-2)(x-5)$ 2
- (d) George has inherited a sum of \$80 000. He deposited it in the bank in an account earning five percent per annum interest compounded annually. He enjoys an annual European holiday and plans to withdraw \$10 000 at the end of each year to finance his holiday.
Let A_n be the amount in the account after n years.
- (i) Show that
$$A_3 = 80000(1.05)^3 - 10000(1.05)^2 - 10000(1.05) - 10000$$
 2
- (ii) Find an expression for A_n .
- (iii) Calculate the number of annual holidays George will be able to fully finance from his account. 3

2013 YEAR 11 MATHS (2U) TASK 1

a) Let $f(x) = x^4 - 3x^2 - 2$.

$$\begin{aligned} f(-x) &= (-x)^4 - 3(-x)^2 - 2 \\ &= x^4 - 3x^2 - 2 \\ &= f(x) \end{aligned} \quad (2)$$

$\therefore f(x) = x^4 - 3x^2 - 2$ is even.

b) $\lim_{x \rightarrow 3} \frac{3x}{x+2} = \frac{9}{5}$ (1)

c) $\sum_{n=1}^6 (2n-3) \Rightarrow$ first term $a = -1$
last term $2(6) - 3 = 9$.

$$\begin{aligned} S_6 &= \frac{n}{2}(a + l) \\ &= \frac{6}{2}(-1 + 9) \\ &= 24. \end{aligned} \quad (2)$$

d) i) $y = x^2 + 2x$ $\frac{dy}{dx} = 2x + 2$ (1)

ii) $y = 3x - 2x^{-3}$ $\frac{dy}{dx} = 3 + 6x^{-4} = 3 + \frac{6}{x^4}$ (2)

iii) $y = \frac{2x}{1+x^2} = \frac{u}{v}$ $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$ (2)
 $= \frac{(1+x^2)(2) - (2x)(2x)}{(1+x^2)^2}$

$$= \frac{2 + 2x^2 - 4x^2}{(1+x^2)^2}$$

$$= \frac{2 - 2x^2}{(1+x^2)^2}$$

$$= \frac{2(1-x^2)}{(1+x^2)^2}$$

$$\frac{-2(x^2-1)}{(x^2+1)^2}$$

e) i) $\frac{a^5}{a^{-3}} = a^8$

①

ii) $\left(\frac{2n^2}{-m}\right)^3 \left(\frac{-m^3}{n}\right)^2$

②

$$= \left(\frac{8n^6}{-m^3}\right) \left(\frac{m^6}{n^2}\right)$$

$$= \frac{8n^6 m^6}{-m^3 n^2}$$

$$= \underline{\underline{-8n^4 m^3}}$$

iii) $\log_3 81 + \log_3 9 + \log_3 243$

①

$$4 + 2 + 5 = 11$$

f) $f(x) = 3x^3 - 3x^2 + bx + 5$

$$f'(x) = 9x^2 - 6x + b$$

②

for only 1 stn $\Delta = 0$

$$\Delta = b^2 - 4ac$$

$$= (-6)^2 - 4(9)(b) = 0$$

$$36 - 36b = 0$$

$$b = 1$$

(2) a) $3x^2 + 9x + 1 = 0$

$$\alpha + \beta = -\frac{b}{a} \quad a = 3, \quad b = 9, \quad c = 1$$

$$\alpha\beta = \frac{c}{a}$$

$$(i) \alpha + \beta = -\frac{9}{3} = -3 \quad (1)$$

$$(ii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-3}{\frac{1}{3}} = -9 \quad (2)$$

$$\begin{aligned} (iii) & (\alpha + 2)(\beta + 2) \\ &= \alpha\beta + 2\alpha + 2\beta + 4 \\ &= \alpha\beta + 2(\alpha + \beta) + 4 \\ &= \frac{1}{3} + 2(-3) + 4 = -\frac{2}{3} \quad (2) \quad \left(-\frac{5}{3}\right) \end{aligned}$$

$$(b) \quad y = \frac{x^2}{2}$$

$$y' = \frac{2x}{2} = x$$

At $x = 4$, gradient tangent is $m_T = 4$

$\therefore$ gradient of normal is $m_N = -\frac{1}{4} \quad (1) \quad (m_1 m_2 = -1)$

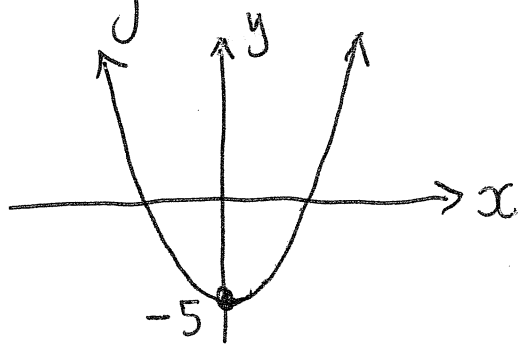
Using $(y - y_1) = m(x - x_1)$

$$y - 8 = -\frac{1}{4}(x - 4)$$

$$y - 8 = -\frac{x}{4} + 1$$

$$y = -\frac{x}{4} + 9 \quad \text{OR} \quad x + 4y - 36 = 0 \quad (2)$$

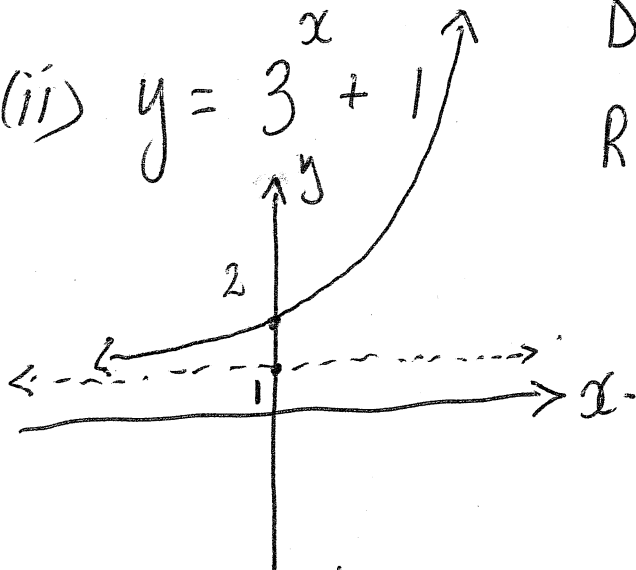
(c) (i) $y = 3x^2 - 5$



$D: \mathbb{R}$ ①

$R: y \geq -5$ ①

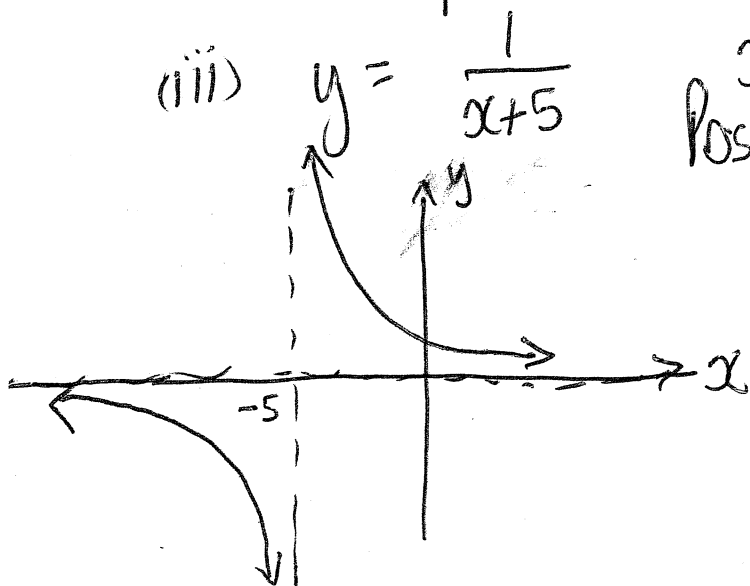
(ii) $y = 3^x + 1$



$D: \mathbb{R}$ ①

$R: y > 1$ ①

(iii) $y = \frac{1}{x+5}$

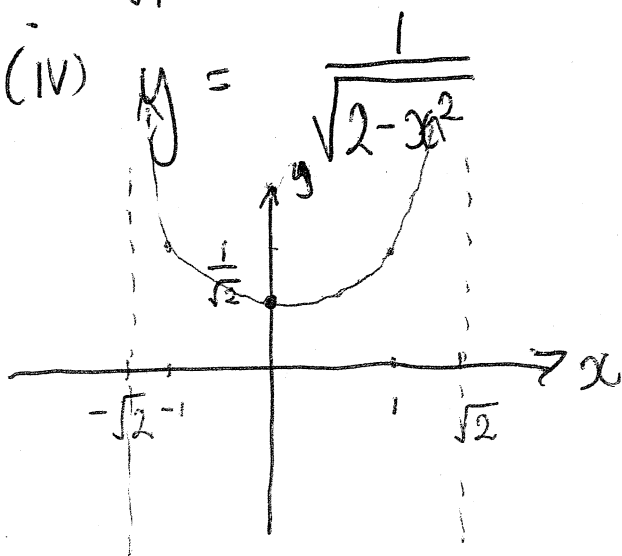


$x \neq -5$
Positive hyperbola.

$D: \text{all } \mathbb{R}, x \neq -5$ ①

$R: \text{all } \mathbb{R}, y \neq 0$ ①

(iv) $y = \frac{1}{\sqrt{2-x^2}}$



$(2-x^2 > 0)$

$D: -\sqrt{2} < x < \sqrt{2}$ ①

$R: y \geq \frac{1}{\sqrt{2}}$ ①

Question 3:

a) $\{1, 2, 3, 4, 5, 6, 6\}$

i) $P(2) = \frac{1}{7}$

ii) $P(6) = \frac{2}{7}$

b) $T_1 = -5, T_2 = -1, T_3 = 3, T_4 = 7$
 $d = 4$

i) $T_n = a + (n-1)d$
 $= -5 + (n-1)4$
 $= -5 + 4n - 4$
 $= 4n - 9$

ii) $S_n = \frac{n}{2} (2a + (n-1)d)$

$S_{20} = \frac{20}{2} (2(-5) + (20-1)4)$
 $= 10(-10 + 76)$
 $= 660$

c) $(x - x_1)^2 + (y - y_1)^2 = r^2$

$\therefore (x - (-3))^2 + (y - 5)^2 = (\sqrt{3})^2$
 $(x + 3)^2 + (y - 5)^2 = 3$

d) $d_1 = \frac{|6x + 8y - 5|}{\sqrt{6^2 + 8^2}}$

$= \frac{|6x + 8y - 5|}{10}$

$d_2 = \frac{|5x + 12y - 1|}{\sqrt{5^2 + 12^2}}$

$= \frac{|5x + 12y - 1|}{13}$

When $d_1 = d_2$

$\frac{|6x + 8y - 5|}{10} = \frac{|5x + 12y - 1|}{13}$

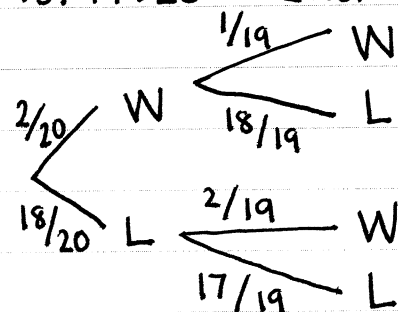
case 1: (Positive)

$13(6x + 8y - 5) = 10(5x + 12y - 1)$
 $78x + 104y - 65 = 50x + 120y - 10$
 $28x - 16y - 55 = 0$

case 2: (one negative)

$78x + 104y - 65 = -50x - 120y + 10$
 $128x + 224y - 75 = 0$

e) 1st Prize 2nd Prize



$$\begin{aligned} \text{i) } P(WW) &= \frac{2}{20} \times \frac{1}{19} \\ &= \frac{1}{190} \end{aligned}$$

ii) At least one

$$\begin{aligned} \therefore 1 - P(LL) \\ &= 1 - \left(\frac{18}{20} \times \frac{17}{19} \right) \\ &= \frac{37}{190} \end{aligned}$$

f) $x^2 + kx + 8 - k = 0$

i) equal roots $\Delta = 0$

$$\begin{aligned} \Delta \Rightarrow k^2 - 4(1)(8 - k) &= 0 \\ k^2 - 32 + 4k &= 0 \\ k^2 + 4k - 32 &= 0 \\ (k - 4)(k + 8) &= 0 \\ \therefore k &= 4, -8 \end{aligned}$$

ii) real and distinct
 $\Delta > 0$

$$\therefore k > 4, k < -8$$

QUESTION 4

$$(a) \sum_{r=1}^{10} r(r+3)$$

$$(b) \quad f(3a-1) = 2(3a-1) - 3 \\ = 6a - 5$$

$$(c) \quad (i) \quad 8^{\frac{1}{2}} \div 2^{\frac{1}{2}} = \left(\frac{8}{2}\right)^{\frac{1}{2}} \\ = 2.$$

$$(ii) \quad \log_a \frac{a^r + a}{a+1} = \log_a a \\ = 1.$$

$$(d) \quad (i) \quad f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x - 1$$

$$f'(x) = x^2 + x - 2$$

$$f''(x) = 2x + 1.$$

For st. points let $f'(x) = 0$

$$(x+2)(x-1) = 0 \\ x = -2, 1$$

$\therefore (1, -\frac{13}{6})$ and $(-2, 2\frac{1}{3})$ are stationary pts.

Test nature $f''(1) = 3 \therefore (1, -\frac{13}{6})$ is a rel. min.
~~min.~~ turning pt.

$f''(-2) = -3 \therefore (-2, 2\frac{1}{3})$ is a rel.
max. turning pt.

Q4 (contd)

(ii) For possible inflexion

$$\text{let } f''(x) = 0$$

$$\text{ie. } 2x + 1 = 0$$

$$x = -\frac{1}{2}$$

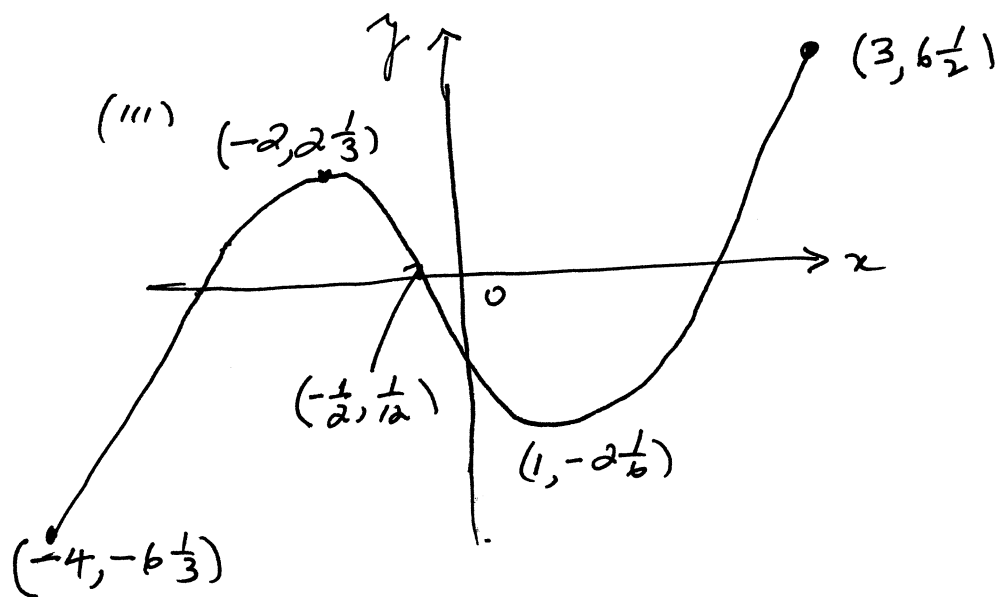
$\therefore (-\frac{1}{2}, \frac{1}{12})$ is a possibility

Test

x	-1	$-\frac{1}{2}$	0
y''	-1	0	1

$\therefore$ change in concavity

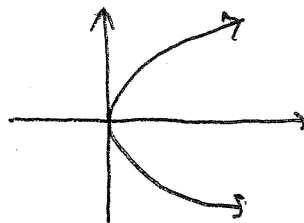
$\therefore (-\frac{1}{2}, \frac{1}{12})$ is a point of inflexion.



(iv) $6\frac{1}{2}$

$$5(a) \quad 3y^2 = 4x$$

$$y^2 = \frac{4}{3}x$$



$$\therefore a = \frac{1}{3} \quad V(0,0)$$

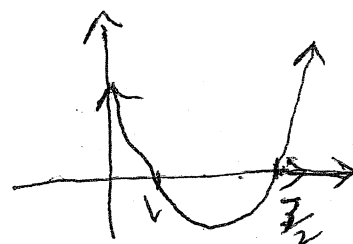
[2]

$\therefore$ Focus $S(\frac{1}{3}, 0)$ Direction $\frac{y}{x} = -\frac{1}{3}$

$$(b) \quad 2x^2 - 9x + 7 > 0$$

$$(2x-7)(x-1) > 0 \quad x = \frac{7}{2}, 1$$

$$x < 1, x > \frac{7}{2}$$



[3]

$$(c) \quad f(x) = x^2 - 6x + 5$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 6(x+h) + 5 - (x^2 - 6x + 5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 6x - 6h + 5 - x^2 + 6x - 5}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 6)}{h}$$

$$= 2x - 6 \quad [3]$$

$$(d)(i) \quad A_0 = 80K$$

$$A_1 = 80K \times 1.05 - 10K$$

$$A_2 = A_1 \times 1.05 - 10K$$

$$= 80K(1.05)^2 - 10K(1.05) - 10K$$

[2]

$$A_3 = 80K(1.05)^3 - 10K(1.05)^2 - 10K(1.05) - 10K$$

$$(ii) \quad A_n = 80K(1.05)^n - \frac{10K(1.05^n - 1)}{1.05 - 1}$$

[3]

(iii) We seek L.N. for $A_n < 0$

$$\text{i.e. } 0 < 80K(1.05)^n(0.05) - 10K(1.05^n - 1)$$

$$0 < 80(1.05)^n(0.05) - 10(1.05^n) + 10$$

$$6(1.05^n) < 10$$

$$n < \frac{\ln 10/6}{\ln 1.05}$$

$$\therefore n < 10.469$$

$\therefore$ 10 holidays are possible.

[3]